SMART INDIA HACKATHON 2026

AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification

- **Problem Statement ID –** SIH26261
- **PROBLEM STATEMENT TITLE-** AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name -** CodeCrew



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AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification

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Problem Statement

Manual Checking Is Slow

- Every application brings many documents, and officers check each one by hand, again and again.
- Missing or expired papers are found late, so files go back and forth and approvals wait.

Fake or Edited Papers Hide Easily

- A changed date or a reused certificate is hard to spot by eye among thousands of files.
- Asking the original issuer to confirm each paper takes a lot of time and effort.

Checks Differ, Proof Is Thin

- Officers may apply the same checklist differently, and findings often lack clear evidence.
- Photos and floor plans are judged by eye, so gaps may only surface in a physical visit.

Proposed Solution

Smart Document Check

- Finds the application type, loads its checklist and reads every document with OCR.
- Flags missing, expired, mismatched or suspicious items so colleges can fix them early.

Verification at the Source

- Checks papers with the original issuer through QR code, digital signature, API or reference number.
- Shows one clear status: Verified, Mismatch, Expired, Pending, Source Unavailable or Manual Review.

Compliance Help and Dashboard

- Compares floor plans and photos with the rules and marks cases needing proof or inspection.
- One dashboard for alerts, findings and a full audit trail. Officers take the final decision.

Uniqueness & Innovation

Evidence on Every Flag

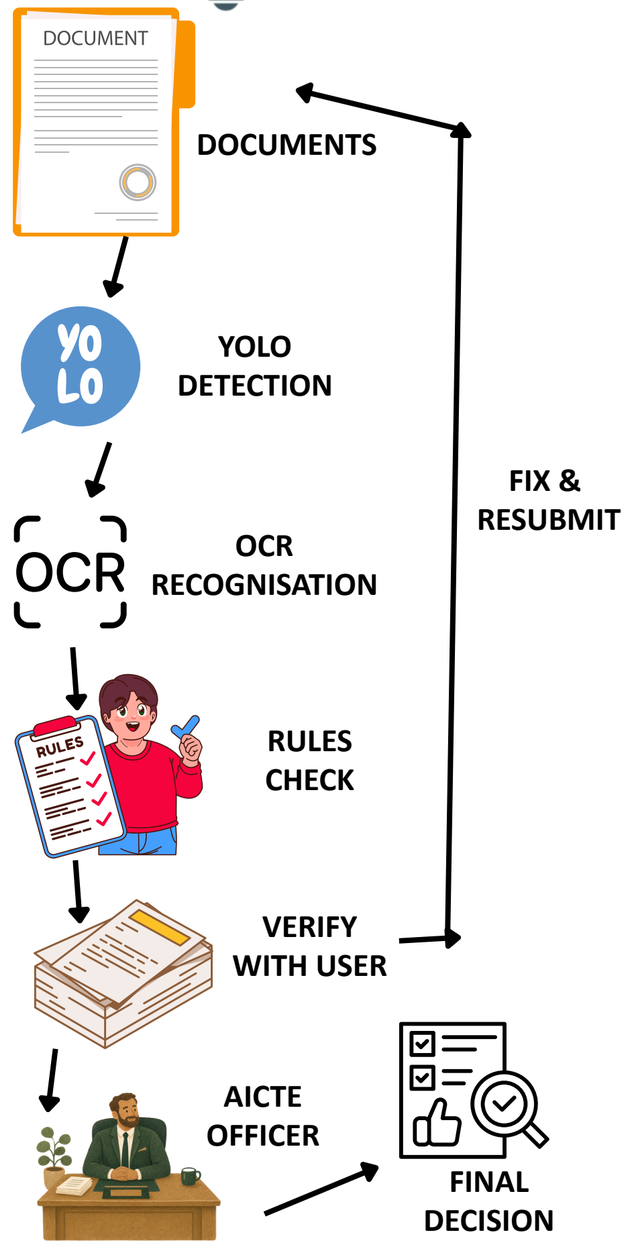
- Each finding points to the exact spot, the rule it broke and the record it conflicts with.
- Rules sit in editable settings linked to the Handbook, so rule changes need no new code.

Layered Fake-Paper Checks

- Signature, QR, file history, image edits and logic checks combine into one risk score.
- Catches the same certificate, photo or faculty reused across colleges.
- Every flag is a hint for human review, never proof of forgery.

Fix Early, Decide Safely

- Colleges run a pre-submission check and fix gaps first, cutting back-and-forth.
- AI only advises and officers decide. Role-based access and a tamper-evident log protect data.



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TECHNICAL APPROACH

Technology Stack

AI DETECTION & OCR DETECTION

YOLO , ROBOFLOW , PADDLE-OCR , OPENCV

FRONTEND

BACKEND

React 18 , Tailwind Recharts

Python fastAPI

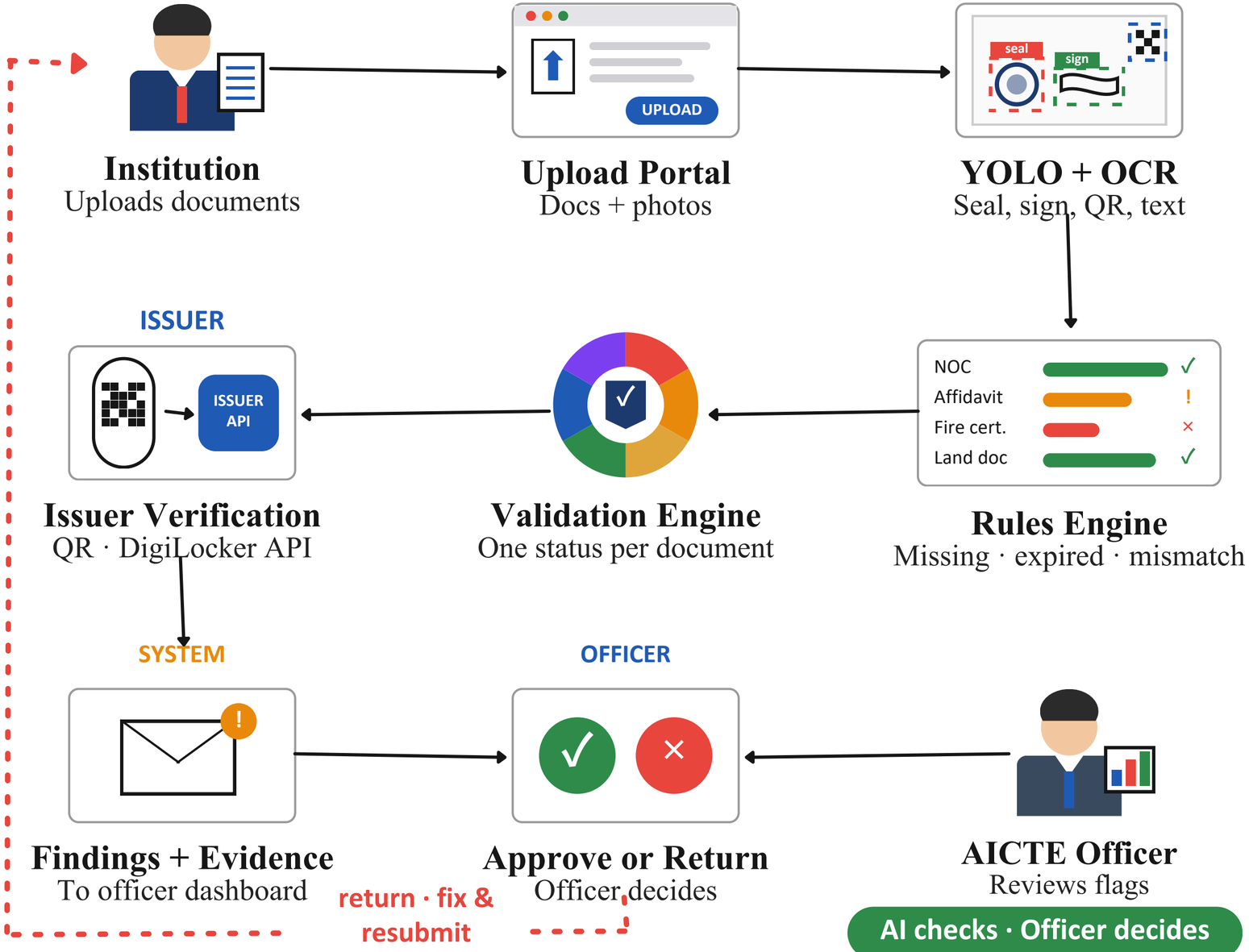
ISSUER VERIFICATION

QR SCAN , DIGILOCKER

DATABASE AND ANOMALY CHECKS

PostgreSQL , EXIFTOOLS , JSON

Core Architecture



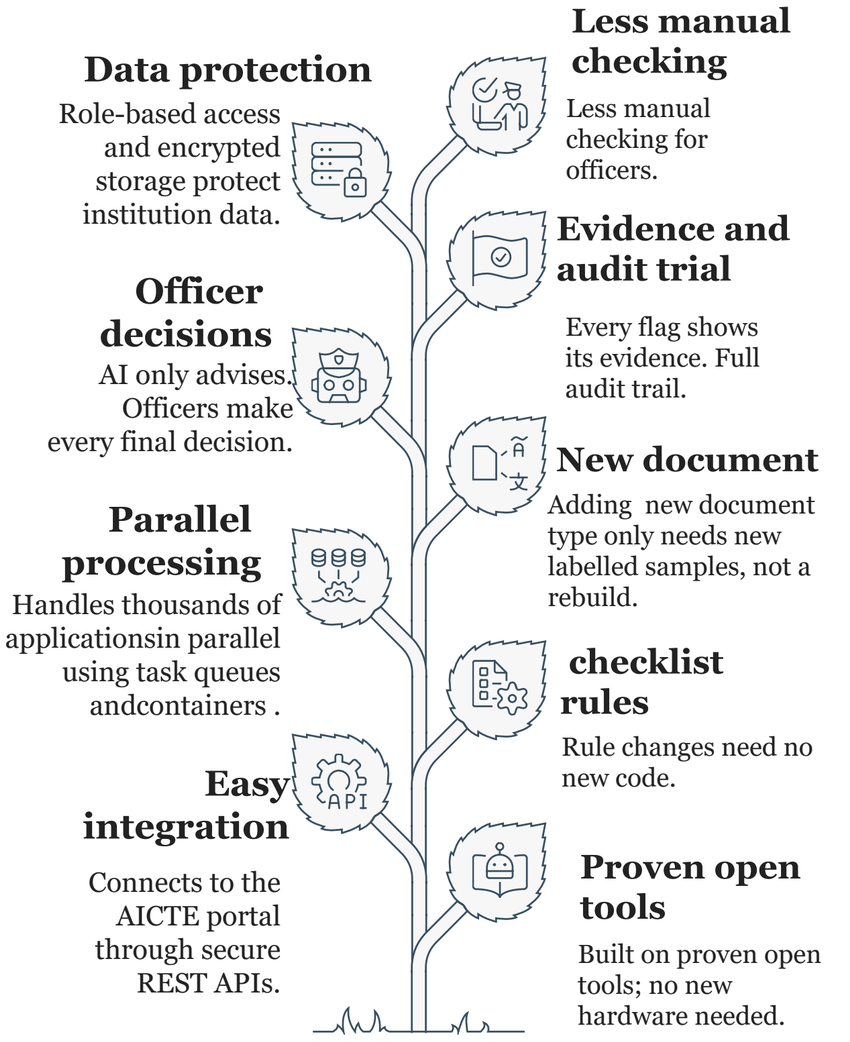
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FEASIBILITY AND VIABILITY

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Delivering Impact at Scale: Feasibility by Design

AICTE platform feasibility at scale



Key Challenges

Slow Manual Checking
Officers read every document by hand, and missing or expired papers are found late, so files go back and forth.

Fake or Edited Documents
A changed date, reused certificate or copied seal is hard to spot by eye, and confirming each paper with the issuer is slow.

Inconsistent Review & Weak Evidence
Officers may apply the checklist differently, and findings often lack clear proof, which slows decisions.

Mitigation Strategies

Smart Pre-Validation
OCR and a Handbook-based rules engine check every document at upload and flag missing, expired or mismatched items instantly.

Detection & Source Verification
YOLO detects seals, signatures and QR codes, anomaly checks catch duplicates and edits, and QR, digital signature or DigiLocker confirm with the issuer.

Evidence-Linked Dashboard
Every flag shows the page, rule and conflicting record. The same rules apply to every file, and the officer decides with a full audit trail.

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IMPACTS AND BENEFITS

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AI Document Validation



Faster Approvals
Validation report is ready at upload, cutting review time.

Fewer Returns
Gaps are fixed before submission, not after rejection.

Fraud Detection
YOLO and anomaly checks flag fake or edited papers.

Source Verification
QR, digital signature and DigiLocker confirm with issuers.

Uniform Rules
Same Handbook rules for every application nationwide.

Transparency
AI explains matching decisions for clarity.

FeedBack Loop
Officer decisions improve detection after each cycle

BENEFITS

Administrative & Economic Benefit

- Cuts officer hours spent on repetitive document checks.
- Flags missing or expired papers at upload, not mid-review.
- Limits costly physical visits to cases that need inspection.

Social Benefit

- Protects students from non-compliant institutions.
- Tells every institution why a document failed, not just "rejected".
- Builds public trust through evidence-backed decisions.

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RESEARCH AND REFERENCES

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ACADEMIC FOUNDATION

- **You Only Look Once: Unified, Real-Time Object Detection** — Redmon et al. (2016)
- **PP-OCR: A Practical Ultra Lightweight OCR System** — Du et al. (2020)
- **LayoutLM: Pre-training of Text and Layout for Document Image Understanding** — Xu et al. (2020)
- **An Overview of the Tesseract OCR Engine** — Smith (2007)
- **A Unified Approach to Interpreting Model Predictions** — SHAP (2017)
- **Perceptual Image Hash Functions** — Zauner (2010), duplicate detection

GOVERNMENT INTEGRATION

- AICTE Portal: **aicte.gov.in**
- Approval Process Handbook 2024-27 — checklist and norms source
- DigiLocker (MeitY) — issuer-verified documents
- National Academic Depository — verified academic records
- IT Act 2000 — legal validity of digital signatures
- DPDP Act 2023 — personal data protection
- NEP 2020 — quality and accountability
- **~29.6 lakh students** in AICTE programmes (2025-26)

TECHNICAL VALIDATION

- **YOLOv8 on Roboflow** — 4 classes: seal, signature, QR, photo
- **mAP@0.5 per class** on a held-out test set
- **PaddleOCR** field extraction checked against ground truth
- **Rules engine** — JSON config per application category
- **6 verification statuses**, Verified to Manual Review
- **Perceptual hashing** for reused documents
- **Evidence + audit log** on every flag
- **Synthetic, anonymized data only**

OUR DELIVERABLES

- GitHub Repository:
- Team: **CodeCrew** · PS **SIH26261**

</> Repository

